DynamicVerse: Physically-Aware Multimodal Modeling for Dynamic 4D Worlds

Anonymous Author(s)
Affiliation
Address
email

Figure 1: The overview of our physically-aware multi-modal annotations framework **DynamicVerse**.

Abstract

Understanding the dynamic physical world, characterized by its evolving 3D structure, real-world motion, and semantic content with textual descriptions, is crucial for human-agent interaction and enables embodied agents to perceive and act within real environments with human-like capabilities. However, existing datasets are often derived from limited simulators or utilize traditional Structure-from-Motion for up-to-scale annotation and offer limited descriptive captioning, which restricts the capacity of foundation models to accurately interpret real-world dynamics from monocular videos, commonly sourced from the internet.

To bridge these gaps, we introduce **DynamicVerse**, a physical-scale, multimodal 4D modeling framework for real-world video. We employ large vision, geometric, and multimodal models to interpret metric-scale static geometry, real-world dynamic motion, instance-level masks, and holistic descriptive captions. By integrating window-based Bundle Adjustment with global optimization, our method converts long real-world video sequences into a comprehensive 4D multimodal format. DynamicVerse delivers a large-scale dataset consists of 100K+ video with

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800K+ annotated masks and 10M+ frames from internet videos. Experimental evaluations on three benchmark tasks—video depth estimation, camera pose estimation, and camera intrinsics estimation—validate that our 4D modeling achieves superior performance in capturing physical-scale measurements with greater global accuracy than existing methods.

1 Introduction

Humans inhabit a dynamic 3D world where geometric structure and semantic content evolve over time, constituting a 4D reality (geometry + time). Understanding this dynamic environment is fundamental for developing advanced AI applications in fields such as robotics [1, 2, 3, 4, 5, 6], extended reality [7, 8, 9], and digital twins [10, 11]. However, building highly generalizable foundation models for these downstream tasks faces a longstanding challenge: acquiring high-quality, ground-truth 4D datasets from real-world environments, given that data-driven solutions increasingly demand 4D data while its collection using multiple sensors remains non-scalable. This raises the question: *Can we develop an automated pipeline capable of generating a real-world 4D dataset at scale?*

Current real-world 4D data primarily focus on indoor scenes [12, 13] or autonomous driving scenarios [14], where geometry capture is straightforward, but their diversity is limited. Even synthetic 4D data [15, 16, 17, 18], while controllable, often lack the fidelity and complexity required to truly represent the real world, resulting in a notable simulation-to-real gap. Moreover, physically-aware multimodal annotations—including metric-scale 3D geometry, detailed representations of non-rigid actors (*e.g.*, object size, mask and bounding box, etc.), and descriptive captions of dynamic contents (*i.e.*, object, camera and scene)—are often absent [19, 20]. This limited data landscape, especially when contrasted with the progress fueled by large-scale datasets in modalities like images, videos, and language, underscores *the compelling need for a large-scale, diverse, physically-aware, and semantically rich annotated multi-modal dataset for 4D scene understanding.*

Against this background, this paper aims to generate scalable, physically-aware, and multimodal annotations from massive monocular video data (see Fig. 1) for numerous potential applications, such as enhancing 4D Vision-Language Models, facilitating advanced long video generation, and enabling linguistic 4D Gaussian Splatting. However, achieving this goal is not trivial. To the best of our knowledge, there is currently a significant lack of rich and diverse 4D datasets (see Tab. 1) adequate for these demanding tasks. To address this data scarcity, we introduce **DynamicGen**, a novel automated data curation pipeline (see Fig. 3) designed to generate physically-aware multi-modal 4D data at scale. This pipeline contains two main stages: (1) metric-scale geometric and moving object recovery (*i.e.*, object category, mask and size) from raw videos, and (2) hierarchical dynamic contents (*i.e.*, object, camera and scene) detailed caption generation. Specifically, the pipeline curates diverse real-world monocular video sources; employs a filtering strategy to remove outliers such as camera motion intensity; integrates multiple foundation models (*i.e.*, VFMs, VLMs, LLMs) for initial frame-wise annotation; applies dynamic bundle adjustment to jointly minimize global photometric error; and concludes with dynamic content captioning at three granularities and human-in-the-loop quality review to ensure annotation semantic accuracy.

The resulting multi-modal 4D dataset, termed **DynamicVerse** (see Fig. 1), comprises over 100K distinct 4D scenes, 800K masklets, and 10M video frames. Each scene is extensively annotated with multiple modalities: metric-scale point maps, camera parameters, object masks with corresponding categories, and detailed descriptive captions. We evaluate DynamicGen through three benchmarks: video depth estimation, camera pose estimation, and camera intrinsics estimation. We demonstrate the generalization capability of DynamicGen to process web-scale video data and extract multi-modal information qualitatively. We also conduct human study and GPT-assited evaluation to validate the quality of generated captions.

Our main contributions are summarized as follows:

- We develop **DynamicGen**, a novel automated data curation pipeline designed to generate physically-aware multi-modal 4D data at scale. This pipeline contains two main stages: (1) metric-scale geometric and moving object recovery from raw videos, and (2) hierarchical detailed semantic captions generation at three granularities (*i.e.*, object, camera and scene). Powered by foundation models (*i.e.*, VFMs, VLMs, LLMs), DynamicGen efficiently generate 4D data at

scale, thus addressing the critical scalability, physical reality and modality diversity limitations of traditional 4D data curation.

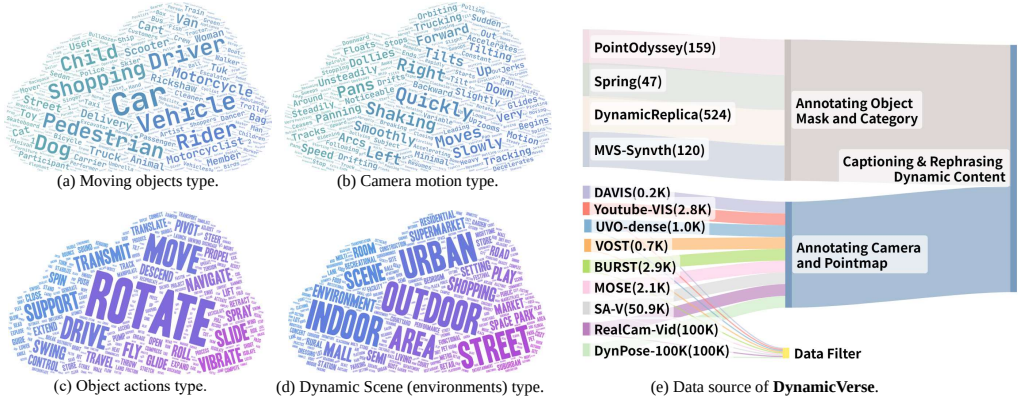
- We introduce **DynamicVerse**, a large-scale 4D dataset featuring diverse dynamic scenes accompanied by rich multi-modal annotations including metric-scale point maps, camera parameters, object masks with corresponding categories, and detailed descriptive captions. DynamicVerse encompasses 100K+ 4D scenes coupled with 800K+ masklets, sourced through a combination of massive 2D video datasets and existing 4D datasets. This represents a significant improvement in terms of data scale, scene and modality diversity compared to prior 4D datasets.
- We validate DynamicGen through three benchmarks: video depth estimation, camera pose and intrinsics estimation. We demonstrate the generalization capability of DynamicGen to process web-scale videos and extract multi-modal information qualitatively. We also conduct human study and GPT-assited evaluation to validate the quality of generated captions.

2 Related Work

Table 1: **Comparison of *DynamicVerse* with large-scale 2D video datasets and existing 4D scene datasets.** DynamicVerse expands the data scale and annotation richness compared to prior works.

	Numerical Statistics			*Provided Annotations*								*Detailed Features*			
Dataset Name	# Videos	# Frames	# Masklets	Camera	Depthmap	Instance Mask	Semantic Mask	Object Category	Object Caption	Scene Caption	Camera Caption	Scene Type	Dynamic Type	Real-world?	Metric-scale?
2D Video Dataset															
DAVIS2017 [21]	0.2K	10.7K	0.4K	✗	✗	✓	✓	✓	✗	✗	✗	-	-	-	-
Youtube-VIS [22]	3.8K	-	8,171	✗	✗	✓	✓	✓	✗	✗	✗	-	-	-	-
UVO-dense [23]	1.0K	68.3K	10.2K	✗	✗	✓	✗	✓	✗	✗	✗	-	-	-	-
VOST [24]	0.7K	75.5K	1.5K	✗	✗	✓	✗	✓	✗	✗	✗	-	-	-	-
BURST [25]	2.9K	195.7K	16.1K	✗	✗	✓	✓	✓	✗	✗	✗	-	-	-	-
MOSE [26]	2.1K	638.8K	5.2K	✗	✗	✓	✗	✓	✗	✗	✗	-	-	-	-
SA-V [27]	50.9K	4.2M	642.6K	✗	✗	✓	✗	✓	✗	✗	✗	-	-	-	-
MiraDATA [28]	330K	-	-	✗	✗	✓	✗	✓	✗	✗	✗	-	-	-	-
4D Scene Dataset															
T.Air Shibuya [29]	7	0.7K	-	✓	✓	✗	✓	✗	✗	✗	✗	Mixed	Street	Synthetic	Yes
MPI Sintel [30]	14	0.7K	-	✓	✗	✓	✗	✓	✗	✗	✗	-	Scripted	Synthetic	-
FlyingThings3D [31]	220	2K	-	✓	✗	✓	✓	✗	✗	✗	✗	Mixed	Objects	Synthetic	-
Waymo [14]	1,150	200K	-	✓	✓	✗	✗	✗	✗	✗	✗	Outdoor	Driving	Real-world	Yes
CoP3D [12]	4,200	600K	-	✓	✗	✓	✗	✗	✗	✗	✗	Mixed	Pets	Real-world	-
Stereo4D [32]	110,000	10,000K	-	✓	✓	✗	✗	✗	✗	✗	✗	Mixed	S. fisheye	Real-world	Yes
PointOdyssey [15]	159	200K	-	✓	✓	✓	✗	✗	✗	✗	✗	Mixed	Realistic	Synthetic	Yes
Spring [16]	47	6K	-	✓	✓	✓	✗	✗	✗	✗	✗	Mixed	Realistic	Synthetic	Yes
Dynamic Replica [17]	524	145K	-	✓	✓	✓	✗	✗	✗	✗	✗	Indoor	Realistic	Synthetic	Yes
MVS-Synth [18]	120	12K	-	✓	✓	✗	✗	✗	✗	✗	✗	Outdoor	Urban	Synthetic	Yes
RealCam-Vid [19]	100K	-	-	✓	✗	✓	✗	✗	✗	✓	✗	Mixed	Realistic	Synthetic	Yes
DynPose-100K [20]	100K	6,806K	-	✓	✗	✗	✗	✗	✗	✗	✗	Mixed	Realistic	Synthetic	Yes
DynamicVerse	100K+	13.6M	800K+	✓	✓	✓	✓	✓	✓	✓	✓	Mixed	Realistic	Real-world	Yes

Multi-modal foundation models. The development of numerous large foundation models in recent years has yielded remarkable performance across multiple tasks such as depth estimation [33, 34, 35, 36, 37], multi-view stereo [1], detection and segmentation [38, 39, 40, 27], human parsing [41], optical flow estimation [1], and point tracking [42, 43, 35]. We propose that these models are highly applicable to achieving holistic 4D understanding, and unifying them within a single framework represents a promising direction for advancing tasks like nonrigid structure from motion. Our DynamicGen pipeline implements this idea by integrating the following pretrained components: UniDepthv2 [44] for geometry initialization, CoTracker3[43] and UniMatch [45] for correspondence initialization, and Qwen2.5-VL [46] and SA2VA [47] for dynamic object segmentation. This integration, coupled with multi-stage optimization and regularization, allows us to extract accurate metric-scale camera poses and 4D geometry from monocular video. Similar to our method, the concurrently developed Uni4D [48] captures 4D geometry and pose, but it suffers from limited data modalities and discontinuous geometric estimates. In contrast, our *DynamicGen* pipeline not only produces globally refined dense 4D geometry but also supports moving object recovery (*i.e.*, object category, mask and size) and provides fine-grained dynamic content (*i.e.*, object, camera and scene) caption annotations.



(a) Moving objects type. (b) Camera motion type. (c) Object actions type. (d) Dynamic Scene (environments) type. (e) Data source of **DynamicVerse**.

Figure 2: The statistics and data source of **DynamicVerse**.

Multi-modal datasets. The development of large-scale multi-modal datasets has proven essential for advancing model performance across numerous domains, including language, image-text (*e.g.*, LAION [49, 50], Conceptual Captions [51], WebImageText [52]), and video understanding (*e.g.*, DAVIS2017 [21], Youtube-VIS [22], UVO-dense [23], VOST [24], BURST [25], MOSE [26], SA-V [27], MiraDATA [28]). Extending this success to holistic 4D understanding requires datasets that capture the dynamic 3D world with rich, multi-modal annotations. Existing 4D datasets, whether from early reconstruction efforts [15, 16, 17, 18] (limited diversity) or recent large-scale posed video collections like RealCam-Vid [19] and DynPose-100K [20] (lacking detailed geometry and semantics beyond pose), and even OBJAVERSE [53] (limited content), fall short of providing the comprehensive multi-modal information needed. Our *DynamicVerse* dataset bridges this gap by offering extensive multi-modal annotations, including metric-scale depth, camera parameters, instance segmentation with labels, and descriptive captions, specifically designed to facilitate advanced 4D research.

3 DynamicVerse

Overview DynamicVerse is a physical-scale, multi-modal 4D modeling framework for real-world video, which contains a novel automated data curation pipeline and corresponding large-scale 4D dataset. The **DynamicGen** pipeline (see Fig. 3) contains two main stages: (1) metric-scale geometric and moving object recovery (*i.e.*, object category, mask and size) from raw videos, and (2) hierarchical dynamic contents (*i.e.*, object, camera and scene) detailed caption generation. This pipeline primarily consists of four steps: 4D scene curation (in Sec. 3.1), data filter strategy (in Sec. 3.2), moving object recovery (in Sec. 3.3), dynamic bundle adjustment (in Sec. 3.4) and dynamic content caption generation (in Sec. 3.5). The resulting **DynamicVerse** dataset comprises over 100K distinct 4D scenes, 800K masklets, and 10M video frames. The data statistics and collection of DynamicVerse are illustrated in Fig. 2.

3.1 4D scene curation

To address the scarcity of available 4D scene data, DynamicGen unifies video data from various real-world video datasets, including DAVIS2017 [21], Youtube-VIS [22], UVO-dense [23], VOST [24], BURST [25], MOSE [26] and SA-V [27], alongside existing synthetic 4D datasets from PointOdyssey [15], Spring [16], Dynamic Replica [17], MVS-Synth [18], RealCam-Vid [19] and DynPose-100K [20]. The inclusion of these datasets is mainly motivated by their potential as scalable data sources for 4D scene understanding.

3.2 Data filter strategy

Data filtering is a critical step for identifying video data suitable for subsequent dynamic bundle adjustment. This process presents challenges due to the noisy quality and inherent variability of video data, which impedes the precise selection of high-quality sequences. To address this, we developed a

Figure 3: The physically-aware multi-modal 4D data generation pipeline **DynamicGen**.

filtering strategy incorporating several distinct criteria: proximal depth, focal-length stability, video blur, camera motion smoothness, and non-perspective distortion. Each of these aspects is quantified by a normalized score. We combine these scores as features and employ a Random Forest model to predict a video quality score ranging from 0 to 5. For model training, we manually annotated approximately 1,000 videos, assigning scores between 0 (indicating largely unsuitable, poor quality or insufficient dynamics) and 5 (indicating highly suitable, good quality and sufficient dynamics). We also employ VLM-judgment to filter unsuitable videos. The detailed data filter process can be found in the Appendix A.1.

3.3 Moving object recovery

To accurately identify the main dynamic objects within a video, we integrated multiple foundation models to achieve reliable segmentation. Specifically, our pipeline first employs Qwen2.5-VL [54] to identify moving objects and determine their semantic categories. These categories are then used to prompt SA2VA [47] for generating corresponding object masks. Leveraging the obtained object masks and categories, we apply physical-aware size extraction to annotate the 3D bounding box for moving objects. Further details are provided in Appendix A.4.

3.4 Dynamic bundle adjustment

Leveraging the high-quality RGB filtered videos, we developed a robust dynamic bundle adjustment method for annotating metric-scale camera parameters and point maps. This task is challenging due to dynamic objects occluding the static scene and static scene appearance changes hindering correspondence estimation. To effectively addresses both difficulties, we design a multi-stage optimization framework, see Fig. 3, including: (1) dynamic masking, (2) coarse camera initialization, (3) tracking-based static area bundle adjustment, (4) tracking-based non-rigid bundle adjustment, and (5) flow-based sliding window global refinement. Compared with traditional Structure-from-Motion techniques [55] and DUSt3R-based methods [56], our framework not only can handle massive video data with different resolutions but also yield metric-scale results by leveraging the full power of various foundation models.

Formulation Given T video RGB frames $\mathbf{I} = (I_1, \ldots, I_T)$ with resolution $H \times W$, we aim to estimate for each timestep $t = 1, \ldots, T$: per-frame pointmap $\mathbf{X}^t \in \mathbb{R}^{H \times W \times 3}$, camera intrinsics $\mathbf{K}^t$, and camera pose $\mathbf{P}^t = [\mathbf{R}^t | \mathbf{T}^t]$, where $\mathbf{R}^t$ and $\mathbf{T}^t$ denote the t-th camera's rotation and translation, respectively. Here, $\mathbf{X}$ contains static points $\mathbf{X}_{static}$ and dynamic points $\mathbf{X}_{dyn}$. We assume all frames share the same intrinsics K where we optimize focal lengths f_x and f_y. The overall cost function is formulated as follows.

$$C_{\text{BA}}(\mathbf{P}, \mathbf{X}_{\text{static}}) + C_{\text{flow}}(\mathbf{X}_{\text{static}}) + C_{\text{NR}}(\mathbf{X}_{\text{dyn}}) + C_{\text{motion}}(\mathbf{X}_{\text{dyn}}) + C_{\text{cam}}(\mathbf{R}) \tag{1}$$

where $C_{\text{BA}}(\mathbf{P}, \mathbf{X}_{\text{static}})$ and $C_{\text{flow}}(\mathbf{X}_{\text{static}})$ are bundle adjustment terms measuring the reprojection error between static correspondences and the static 3D structure $\mathbf{X}_{\text{static}}$. $C_{\text{NR}}(\mathbf{X}_{\text{dyn}})$ is a non-rigid structure-from-motion term evaluating the consistency of the dynamic point cloud with its tracklets. Regularization is applied to camera motion smoothness through $C_{\text{cam}}(\mathbf{P})$ and to the dynamic structure and motion via $C_{\text{motion}}(\mathbf{X}_{\text{dyn}})$. Each term participates in different optimization stages, which are described below. Detailed explanations of the cost terms are provided in the *supplementary material*.

Stage I: Dynamic masking We first extract dynamic masks to filter out the dynamic points for static area bundle adjustment. Specifically, we use semantic-based and motion-based method to obtain dynamic masks $\mathcal{M} = \{\mathbf{M}^t\}_{t=0}^T = \{\mathbf{M}^t_{sem} \cup \mathbf{M}^t_{flow}\}_{t=0}^T$.

For the segmentation-based approach, we use the generated moving object masks $\{\mathbf{M}^t_{seg}\}_{t=0}^T$ in Sec. 3.3. For the flow-based approach, we employ Unimatch [45] to obtain dense optical flow predictions and compute per-frame epipolar error maps, which indicate the likelihood of pixels belonging to the dynamic foreground. Then we can obtain dynamic masks $\mathbf{M}_{flow} = [E_1, E_2, \ldots, E_T]$ by thresholding on these epipolar error maps.

Stage II: Coarse camera initialization In this stage, we start camera initialization by obtaining video depth $\mathcal{D} = \{\mathbf{D}_t\}_{t=0}^T$ and dense pixel motion $\mathcal{Z} = \{\mathbf{Z}_k\}_{k=0}^K$.

For video depth estimation, we use UniDepthV2 [44], a monocular depth estimation network, to estimate initial depth maps $\mathcal{D}$ and initial camera intrinsics $\mathbf{K}_{\text{init}}$. For dense pixel motion estimation, we utilize Co-TrackerV3 [43] for its robustness. We apply Co-Tracker bi-directionally on a dense grid every 10 frames to ensure thorough coverage. We filter and classify tracklets using segmentation masks yielding a set of correspondent point trajectories $\{Z_k \in \mathbb{R}^{T \times 2}\}_{k=0}^K$ at visible time steps determined by Co-Tracker. Combining $\mathcal{D}$ and $\mathcal{Z}$ allows us to establish 2D-to-3D correspondences. This allows us to initialize and tune camera parameter $\mathbf{P}$ by minimizing the following cost function with respect to camera parameters *only*. Specifically, we can unproject each video frame's depth at time t back to 3D and minimize the following cost function:

$$\min_{\mathbf{P}} \sum_{(t',t)} \sum_{\mathbf{z}_k \in \neg\mathcal{M}} \|\mathbf{z}_{k,t'} - \pi_{\mathbf{K}}(\pi_{\mathbf{K}}^{-1}(\mathbf{z}_{k,t}, \mathbf{D}_t, \boldsymbol{\xi}_t), \boldsymbol{\xi}_{t'})\|_2^2 \tag{1}$$

where $\pi_{\mathbf{K}}^{-1}$ is the unprojection function that maps 2D coordinates into 3D world coordinates using estimated depth $\mathbf{D}_t$. We perform this over all pairs within a temporal sliding window of 5 frames. Given camera initialization $\hat{\mathbf{P}}$, we unproject our depth prediction into a common world coordinate system, which provides an initial 4D structure $\hat{\mathbf{X}}$. This is used as initialization for later optimization.

Stage III: Static area bundle adjustment We jointly optimizes camera pose and static geometry by minimizing the static component-related energy in a bundle adjustment fashion. Formally speaking, we solve the following:

$$\min_{\mathbf{P}, \mathbf{X}_{\text{static}}} C_{\text{BA}}(\mathbf{P}, \mathbf{X}_{\text{static}}; \mathcal{Z}, \mathcal{M}) + C_{\text{cam}}(\mathbf{R}) \tag{2}$$

By enforcing consistency with each other, this improves both the static geometry and the camera pose quality. We perform a final scene integration by unprojecting correspondences into 3D using improved pose and filtering outlier noisy points in 3D.

Stage IV: Non-rigid bundle adjustment Given the estimated camera pose, this stage focuses on inferring dynamic structure. Note that we freeze camera parameters in this stage, as we find that incorrect geometry and motion evidence often harm camera pose estimation rather than improve it.

Additionally, enabling camera pose optimization introduces extra flexibility in this ill-posed problem, harming robustness. Formally speaking, we solve the following:

$$\min_{\mathbf{X}_{\mathrm{dyn}}} C_{\mathrm{NR}}(\mathbf{X}_{\mathrm{dyn}}; \mathbf{P}, \mathcal{Z}, \mathcal{M}) + C_{\mathrm{motion}}(\mathbf{X}_{\mathrm{dyn}}) \tag{3}$$

We initialize $\mathbf{X}_{\mathrm{dyn}}$ using video depth and our optimized camera pose from last step. This energy optimization might still leave some high-energy noisy points, often from incorrect cues, motion boundaries, or occlusions. We filter these outliers based on their energy values in a final step. To further densify the global point cloud, enabling each pixel to correspond to a 3D point, we perform depthbased interpolation by computing a scale offset.

Stage V: Sliding window global refinement Given the estimated optical flow, this stage focuses on refining static structure. Note that we freeze camera parameters in this stage. Formally speaking, we solve the following:

$$\min_{\mathbf{P}_{\mathrm{static}}} E_{\mathrm{flow}}(\mathbf{P}_{\mathrm{static}}) \tag{4}$$

With consideration for accuracy and efficiency, the sliding window global refinement is capable of significantly enhancing the multi-view consistency of static points and generalizing effectively to real-world 4D scenes. The detailed process can be found in the *supplementary material*.

3.5 Dynamic Content Caption Generation

Drawing upon the emphasis placed by LEO [57] and SceneVerse [58] on the criticality of caption quality and granularity for comprehensive scene understanding, we design captions at three specific levels: object, scene, and camera. Object captioning focuses on detailed object motion, scene captioning describes object-scene interactions, and camera captioning conveys intricate camera movement. To augment the caption, Large Language Models (LLMs) are employed to automatically rephrase initial captions and align them with these three granularity levels. Finally, to ensure data quality, human verification is conducted to filter out low-quality caption annotations.

Moving object captioning. Moving object captions provide detailed descriptions crucial for object grounding. However, prior datasets often have incorrect temporal alignment [58] or insufficient detail [15, 59], while current video captioning methods yield only simple (*e.g.*, Panda-70M [60]) or non-localized descriptions (*e.g.*, Qwen2.5-VL [54]). To address these limitations and generate detailed, accurate captions for individual objects, we utilize DAM [61], known for its superior capabilities. Given RGB video and object masks, DAM [61] generates these descriptions using designed prompts. The detailed pipeline is in the supplementary material.

Dynamic scene captioning. Scene-level captions are designed to capture global information, depicting the key objects within the scene along with their associated actions, interactions, and functionalities. For a comprehensive understanding of the entire dynamic scene, we utilize Qwen2.5-VL [54] for dynamic scene captioning. To obtain more detailed, fine-grained, and accurate captions, we propose the use of structured captions. This process involves leveraging the fine-grained moving object captions as auxiliary input and employing specific prompting to generate the final scene-level descriptions. In the design of the prompts, we discovered that an explicit Hierarchical Prompt Design significantly aids the Qwen2.5-VL[54] in comprehending its role, its expected format, and its operational boundaries. This approach contributes to the stabilization of the output's format and enhances the overall quality of the results. For more details, please refer to *supplementary material*.

Camera motion captioning. Camera Motion Captioning aims to describe the camera's trajectory and movement patterns. Using the powerful VLM, we analyze the sequence of inter-frame transformations to identify key motion types like panning, tilting, zooming, and dolly movements. This kinematic information is then used to generate natural language descriptions, potentially leveraging template-based generation or LLM prompting, to convey how the viewpoint changes over time.

Caption rephrasing. Following the generation of the three distinct caption types—object, scene, and camera motion—a Large Language Model (LLM) [54] is employed for their joint processing. This collective processing is performed to align the dynamic content descriptions across the various caption types and to rephrase them, thereby enhancing both consistency and readability.

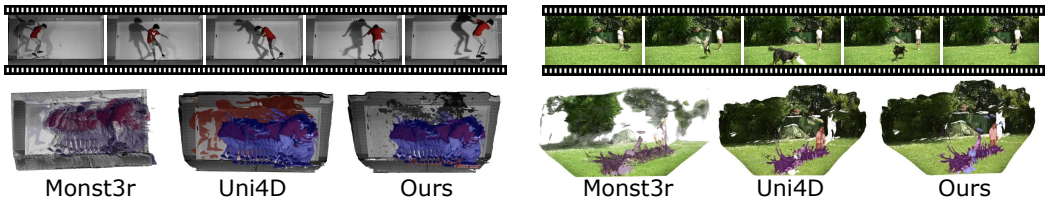


Figure 4: **Qualitative results of 4D reconstruction on in-the-wild data.**

Human-in-the-loop quality review. To provide a faithful comparison against larger pretrained models, human evaluation was used. Addressing persistent errors from source annotation inaccuracies, we implemented an iterative human-in-the-loop verification during caption construction to identify errors, trace sources, and revise/remove problematic data

4 Experiments

In this section, we present experimental results to evaluate the robustness of our *DynamicGen* pipeline. Due to the page limit, we direct readers to the *supplementary* for implementation details, more qualitative results, and more experimental analyses.

Table 2: Video depth evaluation on Sintel and KITTI datasets. **Bold** and underlined values indicate best and second best results.

			Sintel		KITTI	
Alignment	Category	Method	Abs↓	δ1.25↑	Abs↓	δ1.25↑
Per-sequence scale	Joint depth & pose	Monst3r [56]	0.344	55.9	0.089	91.4
		Uni4D [48]	**0.289**	**64.9**	**0.086**	**93.3**
Per-sequence scale & shift	Single-frame depth	Depth-pro [33]	0.280	60.5	0.080	94.2
		Metric3D [62]	**0.205**	**71.9**	**0.039**	**98.8**
	Video depth	DepthCrafter [34]	0.231	69.0	0.112	88.4
	Joint video depth & pose	Robust-CVD [63]	0.358	49.7	0.182	72.9
		CasualSAM [64]	0.292	56.9	0.113	88.3
		Uni4D [48]	0.216	72.5	0.098	89.7
		***DynamicGen*(Ours)**	**0.215**	**73.2**	**0.091**	**91.2**

4.1 Video Depth Estimation

To evaluate video depth estimation accuracy, we assess several baseline methods, including metric depth predictors such as Metric3Dv2 [62], Depth-Pro [33], DepthCrafter [34], and Unidepth [36], which operate without scale or shift alignment. We also consider joint 4D modeling approaches, including MonST3R [56] and RCVD [63]. Evaluations are conducted on the Sintel [30], Bonn [65], and KITTI [66] datasets, following standard protocols [34] by applying global shift and scale alignment to the predicted depth maps. We report absolute relative error (Abs Rel) and the percentage of inlier points ($\delta < 1.25$), with all methods undergoing least-squares alignment in disparity space. As shown in Tab. 2, *DynamicGen* achieves the best overall performance across all datasets and evaluation metrics. In particular, it consistently outperforms prior approaches in both absolute accuracy and geometric consistency, demonstrating strong generalization to diverse and dynamic scenes.

4.2 Camera Pose Estimation

We evaluate our method against recent dynamic scene pose estimation approaches, including learning-based visual odometry (e.g., LEAP-VO [67], DPVO [68]) and joint depth-pose optimization methods (e.g., Robust-CVD [63], CasualSAM [64], MonST3R [56]). Experiments are conducted on the Sintel [30] and TUM-dynamics [69] datasets, following LEAP-VO's split for Sintel and subsampling the first 270 frames of TUM-dynamics, as done in MonST3R. Camera trajectories are aligned using Umeyama alignment [70], and we report Absolute Trajectory Error (ATE), Relative Translation Error

(RPE trans), and Relative Rotation Error (RPE rot). As shown in Tab. 3, *DynamicGen* consistently achieves state-of-the-art results across all metrics and datasets, outperforming existing methods in both translation and rotation accuracy.

Table 3: Camera Pose Evaluation on Sintel and TUM-dynamic datasets. **Bold** and underlined values indicate best and second best results.

		Sintel			TUM-dynamics		
Category	Method	ATE↓	RPE trans↓	RPE rot↓	ATE↓	RPE trans↓	RPE rot↓
Pose only	DPVO [68]	0.171	**0.063**	**1.291**	**0.019**	**0.014**	**0.406**
	LEAP-VO [67]	**0.035**	0.065	1.669	0.025	0.031	2.843
Joint depth & pose	Robust-CVD [63]	0.368	0.153	3.462	0.096	0.027	2.590
	CasualSAM [64]	0.137	0.039	0.630	0.036	0.018	0.745
	Monst3r [56]	**0.108**	0.043	0.729	0.108	0.022	1.371
	Uni4D [48]	0.110	0.032	0.338	**0.012**	**0.004**	0.335
	***DynamicGen*(Ours)**	**0.108**	**0.031**	**0.337**	**0.012**	**0.004**	**0.331**

4.3 Camera Intrinsics Estimation

Camera intrinsics are typically unavailable for most casual videos, especially those sourced from the Internet. However, accurate intrinsics are critical for reliable pose estimation and 3D reconstruction. To assess this, we evaluate focal length estimation accuracy on the Sintel dataset, with results summarized in Tab. 4. UniDepth predicts depth and focal length from a single image, while Dust3r processes sequential frames but is trained under classical multi-view settings and fails to generalize well to dynamic scenes. In contrast, *DynamicGen* demonstrates strong generalization to dynamic content and achieves the best performance in both Absolute Focal Error (AFE) and Relative Focal Error (RFE), setting a new state of the art for focal length estimation in unconstrained video scenarios.

Table 4: **Camera intrinsics estimation.**

Method	$AFE_{(px)}$↓	$RFE_{(\%)}$↓
UniDepth [36]	447.4	0.357
Dust3r [71]	434.0	0.364
***DynamicGen*(Ours)**	**413.1**	**0.241**

Table 5: **Dynamic Scene Caption evaluation.**

Method	Acc.↑	Com.↑	Con.↑	Rel.↑	Avg.↑
DO	83.93	81.47	51.58	69.69	71.67
COT	**86.77**	**85.13**	**55.94**	**76.52**	**76.09**

4.4 Caption Quality Evaluation

We evaluate caption quality by randomly collecting 500 samples from SA-V [27]. The experimental results presented in Tab. 5 strongly suggest that incorporating a Chain-of-Thought prompting strategy significantly improves the quality of dynamic scene captions generated by VLMs, as evaluated by the LLM-as-Judge metric G-VEval [72]. COT leads to captions that are more accurate, complete, concise, and relevant compared to direct output (DO), indicating the effectiveness of using intermediate reasoning steps for this task.

5 Conclusion

In this work, we addressed the critical limitations in traditional 4D data curation concerning scalability, physical reality, and modality diversity. We introduced DynamicGen, a novel automated pipeline leveraging foundation models for metric-scale geometric and moving object recovery, alongside hierarchical detailed semantic captioning from raw videos. We rigorously validated the capabilities of DynamicGen through standard benchmarks for video depth and camera pose/intrinsics estimation, qualitative generalization analysis on diverse web videos, and human/LLM-assisted evaluations confirming the high quality of the generated captions Utilizing DynamicGen, we successfully constructed DynamicVerse, a large-scale 4D dataset offering over 100K dynamic scenes with rich physically-aware multi-modal annotations. Collectively, this work provides both a robust and scalable methodology for 4D data generation and a comprehensive new resource, DynamicVerse, to drive future research in dynamic 4D scene understanding.

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A Appendix

In the appendix, we provide more results and analysis and summarize them as follows:

- In Section A.1, we detail the video data filter process.
- In Section A.2, we introduce the hierarchical design of dynamic content caption generation.
- In Section A.3, we present the template of our hierarchical prompt.
- In Section A.4, we present the detailed process of physical-aware object size extraction.
- In Section A.5, we provide the detailed pseudo code of semantic-aware key-frame extraction, outlier removal with DBSCAN, and canonicalization algorithm.

A.1 Video Data Filtering.

The goal of video data filtering is to identify videos that contain motions such as object motion, camera motion, interaction between objects, human actions, sequences of actions, and manipulation of objects, while rejecting videos with static scenes, like landscapes, or those that are artificial or highly edited.

We start by defining specific criteria to identify videos suitable for dynamic bundle adjustment, which we detect via subsequently defined data filtering steps. We argue that videos should satisfy three criteria:

C1. **Real-world and quality video**. Videos must be from the real world i.e. not cartoons, computer screen recordings, side-by-side/composited videos, and heavily edited processed content. The videos should also be of sufficient quality, resolution, frame rate and should not have non-perspective distortion.

C2. **Feasibility for pose prediction**. Videos should not contain severe zoom-in or zoom-out effects, abrupt shot changes, or ambiguous reference frames (e.g. scenes filmed inside a moving car). Videos without static correspondences, e.g., when the background is fully blurredor occluded should be excluded.

C3. **Dynamic camera and scene**. Non-static camera allows non-trivial pose for *e.g.* training pose estimation or camera-controlled video generation methods. Videos with moving cameras and dynamic scenes typically also provide richer data, *e.g.*, interactions involving people, supporting a wider range of downstream applications.

To achieve these, we created a novel video filtering pipeline consisting of the following steps:

S1. **Compute motion features**. For each video, we compute a list of features from video frames, including video blur degree, frames per second (fps), number of frames, number of I-frames, motion vector magnitude, and motion vector variance, using off-the-shelf tools like OpenCV [10].

S2. **Extract video frame features.**. For each video, we employ *semantic-aware key frame extraction* and encode these keyframes using a DINOv2 model [98] and a SigLIP model [160]. The pseudo-code of *semantic-aware key frame extraction* can be found here in Fig. 9.

S2. **Obtain geometric features.**. For each video, we employ VGGT and UniDepth-V2 to obtain camera and geometry. Then we employ proximal depth verification, focal-length stability and camera motion smoothness assessment.

S3. **LLM Features**. For each video, we also run a multimodal large language model (LLM) like Qwen2.5-VL [54] to extract MLLM features. We composed a list of 26 questions and performed MLLM inference on the videos. The questions can be found in Fig. 5.

S4. **Video Quality Scoring**. We combine all the features collected so far and use a random forest model to predict a score between 0 and 5. To train the model, we manually annotated approximately 1,000 videos with scores between 0 and 5. A low score indicates that the video is almost static and can be nearly summarized by a single frame, while a high score indicates that there are multiple temporal events in the video, requiring several frames to accurately caption it. We use these annotated videos as training data to fit a random forest model for video quality score prediction.

S5. **VLM Assistence**. After filtering based on video quality score, we used VLM to further filter the remain videos.

LLM Feature extraction question list

Quesiton List
Is the camera capturing the scene static? Reply yes or no.
Is the camera capturing the scene moving? Reply yes or no.
Is the video capturing a landscape? Reply yes or no.
Is the video capturing a static scene? Reply yes or no.
Is the scene captured from a distance? Reply yes or no.
Is the video captured with a drone? Reply yes or no.
Is the video computer-generated? Reply yes or no.
Is the video content abstract? Reply yes or no.
Is there something moving through the scene? Reply yes or no.
Is there someone doing something in the video? Reply yes or no.
Are there several things moving in the video? Reply yes or no.
Is there an object that is being manipulated? Reply yes or no.
Are there animals in the video? Reply yes or no.
Is the scene mostly static? Reply yes or no.
Are things occluding each other in this video? Reply yes or no.
Is there something obstructing the view apart from the watermark? Reply yes or no.
Is there a large number of things in the video? Reply yes or no.
Are there more than 5 different objects in the video? Reply yes or no.
Is it hard to keep track of some entities because they are moving so much? Reply yes or no.
Is someone looking at a phone, a tablet or a computer screen? Reply yes or no.
Are they looking at a phone, a tablet or a computer screen during the whole video? Reply yes or no.
Are there several moving persons in this video? Reply yes or no.
Are there several moving animals in this video? Reply yes or no.
Are there several objects in this video? Reply yes or no.
Are there several similar-looking objects in the video? Reply yes or no. Do they look similar? Reply yes or no.

Figure 5: LLM Feature Extraction.

A.2 Hierarchical Prompts for Dynamic Content Captioning.

This design is introduced to enhance the ability of multi-modal and language models to effectively perform their roles during the captioning process. We illustrate the caption prompt components in Figs. 6 and 7, and the summary prompt in Fig. 8. Our hierarchical prompts primarily consist of four components: Character, Tasks, Constraints, and Structured Input. The Character component defines the model's role and the operational context it is expected to handle. The Tasks section specifies the required capabilities and objectives the model must fulfill, ensuring accurate and comprehensive execution. The Constraints section clarifies prohibited behaviors and rules governing the structure or content of the output. The Structured Input component defines the required format for input data, which should be configured according to the specific scenario. For example, when generating dynamic object captions, the Character component instructs the model to act as an expert in analyzing video frames. The Tasks section requires the model to describe the scene by addressing several aspects, including (1) summarizing object behavior, (2) describing visual and physical details, and (3) tracking temporal changes. The Constraints section mandates precise descriptions and prohibits itemized lists in the output. In the Structured Input section, the input typically includes different video frames extracted from semantic-aware key-frame extraction.

A.3 Hierarchical Prompt Template.

The hierarchical prompt templates of object captioning, scene captioning and caption rephrasing are shown as follows.

A.4 Physical-aware Size Extraction.

2D Object to 3D Object Lifting As we mentioned in method section 3.3, we use a variety of off-the-shelf models to extract relavent information: object categories and masks. We then run the state-of-the-art metric depth detector, UniDepthV2 [44] on the image. UniDepthV2 outputs metric

The Prompt Temple of Object Captioning

 # Character
You are an excellent video frame analyst. Given the video in the form of a sequence of frames above, describe the object in the masked region in the video and the following task requirements and constraints need to be met.

 # Tasks
- ## Task1: Summarizing Object Behavior
- Provide an overall summary of the masked object's actions and movements.
- Highlight key interactions, motion patterns, and transitions.

- ## Task2: Describing Visual and Physical Details
- Describe the masked object's visual features (e.g., shape, color, texture) and physical properties (e.g., material, etc)
- Note consistency or variations in these attributes within the masked region.

- ## Task3: Tracking Temporal Changes
- Systematically explain changes in the masked object's behavior, appearance, or motion over time.
- Focus on observable shifts in actions, attributes, or interactions within the sequence of frames.

 # Constraints
- You should express it in a coherent sentence.
- State facts objectively without using any rhetorical devices such as metaphors or personification.
- Stick to a narrative format for descriptions, avoiding list-like itemizations.
- Descriptions should be fluent and precise, avoiding analyzing and waxing lyrical.
- Descriptions need to be concise, describing only the information that can be determined, without analysis orspeculation.
- Do not mention the frame number and timestamp of the current frame.
- Description should be rich and precise enough for an AI to accurately reconstruct the object's movements

 # Structured Input
Video: <image><image><image><image><image><image><image><image>

Figure 6: The Prompt Temple of Dynamic Object Caption.

depth (in real-world "meters"). Combined with an fov estimation, we are able to to lift 2D images into 3D point clouds as illustrated with real-world scale

In this point cloud processing step, outliers, or points that significantly deviate from the main group, are removed to enhance data accuracy. A clustering algorithm DBSCAN [73] groups the points based on proximity, focusing on densely populated regions and eliminating sparse, less significant points. This results in a cleaner, more structured point cloud, ideal for subsequent shape and geometry analysis where dimensions of the shapes are measured. Since we already obtained the semantic segmentation for the point cloud, we may use this information to process outliers at adaptive scales. For smaller objects, we use a smaller threshold, proportional to the along each axis. We observe that such choice effectively removes point cloud outliers while also keeping important points for smaller objects. We provide pseudo code below in Fig. 10.

Coordinate Canonicalization Now we have a 3D point cloud under metric scale. However, the point cloud is still in camera frame, which limits the information we can extract. For example, an object closer to the upper side of the image isn't necessarily further from the ground, because the camera might be pointing at the ground instead of the front. In order to solve this problem, we canonicalize the coordinate system of the pointcloud by detecting horizontal surfaces. We use a light weight segmentation model [74] to segment out pixels correspond to categories like "floor", "table top", before using RANSAC to fit the biggest plane among these 3D points.

When we detect a surface defined by enough points, we canonicalize the coordinate of the point cloud by creating a new origin by projecting camera origin to the detected plane. We use the normal axis of the detected plane as z-axis and project the original z-axis of camera on the the plane as the new x-axis. By doing so, when we can detect horizontal surfaces like ground, we effectively transform the point cloud into world coordinate instead of camera coordinate. A more detailed algorithm can be found in Fig. 11. On the other hand, when not enough points corresponding to horizontal surfaces are detected, we flag canonicalization as failed and avoid synthesizing questions that depends on canonicalization, such as questions about elevation.

The Prompt Temple of Scene Captioning

 # Character
You are a spatial-semantic video analyst specializing in reconstructing dynamic scene graphs. You have an understanding of the overall motion process for every dynamic object: '{object_caption}'. Now, your expertise lies in describing the scene, parsing object hierarchies, quantifying spatial relationships, and tracing motion patterns to build dynamic scene captions and the following task requirements and constraints need to be met.

 # Tasks

- ## Task1:Scene Context Decomposition
- Classify scene type (indoor/outdoor/functional space)
- Describe background: lighting conditions, dominant environmental elements.
- Calculate objects count per hierarchical tier.

- ## Task2: Object Hierarchy Categorization
- Tier 1: Primary interactive agents (humans, vehicles)
- Tier 2: Functional complements (tools, furniture)
- Tier 3: Static environment (walls, terrain)
- For each object analyze:
- Class (semantic label): e.g., "coffee mug".
- Location: spatial position within the entire scene.
- Size: relative size of the scene.
- Orientation: Cardinal/relative facing direction.

- ## Task3: Spatial Relationship Mapping
- Define scene graph edges using:
- Proximity: Adjacent/nearby/distant
- Directional: Left-right/front-behind/above-below
- Composite groups: Clustered objects (e.g., "dishwasher containing plates")
- Highlight 2-3 critical relationships shaping scene logic.

- ## Task4: Motion Pattern Analysis
- For dynamic objects:
- Trace trajectory patterns (linear/curved/zigzag).
- Speed: Accelerating/decelerating/constant.
- Map interaction zones: Areas where motion paths intersect.

- ## Task5: Scenario Reasoning
- Synthesize your observations to infer plausible human activities (e.g., food preparation in a kitchen), scene functionality, and aesthetic style, ensuring all inferences strictly adhere to observable evidence.
- Focus on meaning derived from dynamic interactions—for instance, a bicycle moving toward an intersection suggests urban commuting, while a cluster of chairs facing a screen implies a meeting space.

 # Constraints
- You should express it in a coherent sentence.
- Use strict geometric terminology: Avoid metaphorical descriptions
- Maintain object permanence: Track identical objects across frames
- Disclose limitations: e.g., "Hand-object occlusion prevents interaction verification"
- Synthesize findings into single 500-word narrative without markdown
- All inferences must derive exclusively from observable spatial-motion data
- Description should be rich and precise enough for an AI to accurately reconstruct the video
- Do not add extra fields or omit any required elements

 # Structured Input
Video: <image><image><image><image><image><image><image><image>

Figure 7: The Prompt Temple of Dynamic Scene Caption.

A.5 Pseudo Code.

The Prompt Temple of Caption Rephrasing

 # Character

You are an expert caption-refiner. scene caption: {scene_caption}, camera motion caption: {camera_caption}, object caption: {object_caption}. Now, you receive the above three captions describing the same video from different perspectives. Your goal is to produce a single, internally consistent set of aligned captions and the following task requirements need to be met.

 # Tasks

- Use the same name or description for the object across all captions, replacing any abstract placeholders in the camera-motion caption with the concrete object term used elsewhere.
- Do not introduce any new entities or relationships beyond those present in the input captions.
- Identify and correct any mistakes or nonsensical phrases (typos, grammatical errors, or semantic inconsistencies).
- Polish wording and enrich expression without changing meaning.
- Three refined captions should be given respectively
- The final caption after polishing should not be less than the original caption
- Don't refine the content, just optimize the vocabulary and sentence structure

 # Structured Input

Text: <object caption><scene caption><camera caption>

Figure 8: The Prompt Temple of Caption Rephrasing.

```
# Input:
#   video,           - Video to be processed
#   time_interval,   - Minimum time interval between keyframes (in seconds)
#   threshold        - Inter-frame similarity threshold

# Output:
#   key_frame_pool - Extracted keyframes

# Calculate minimum frame interval based on video frame rate
frame_interval = time_interval * video.fps

# Get all frames from the video
video_frame_list = video.frame_list

# Initialize keyframe pool with the first frame
key_frame_pool.append(video_frame_list[0])

# Semantic-aware keyframe extract
# Iterate through video frames with the specified interval, starting from the second frame
for idx in range(1, len(video_frame_list), frame_interval):
    # Get semantic features for the last keyframe added and the current frame being considered
    frame_feature_1 = get_clip_cls_token(key_frame_pool[-1])
    frame_feature_2 = get_clip_cls_token(video_frame_list[idx])

    # Calculate similarity between the features
    similarity = similarity(frame_feature_1, frame_feature_2)

    # If similarity is below the threshold, indicating a significant change, add the current frame as a
keyframe
    if similarity < threshold:
        key_frame_pool.append(video_frame_list[idx])

# Accommodate videos with small content changes
# If only the first frame was selected (implying little change), add the last frame as well
if len(key_frame_pool) == 1:
    key_frame_pool.append(video_frame_list[-1])

# Note: The key_frame_pool now contains the extracted keyframes.
```

Figure 9: Pseudo code for semantic-aware key-frame extraction.

```
# Input:
# points_obj: Pointcloud of object of interest
# pcd: Full Pointcloud

# Calculate scale factor from object stddev for parameter scaling.
scale = norm(points_obj.std(axis=0)) * 3.0 + 1e-6

# Remove statistical outliers based on distance to neighbors.
pcd = pcd.remove_stat_outlier(neighbors=50, std=1.2)

# Downsample pointcloud using voxel grid; voxel size is scaled.
pcd = pcd.down_sample(voxel_size=max(0.01, scale / 20))

# Perform DBSCAN clustering. Unclustered points are labeled as noise/outliers.
# eps: Neighborhood search radius (scaled).
# min_points: Minimum points for a cluster (based on total points).
labels = array(pcd.cluster_dbscan(
    eps=scale / 3.6, min_points=len(pcd) // 10))

# Points with label -1 from DBSCAN are considered noise/outliers.
```

Figure 10: Pseudo code for outlier removal with DBSCAN.

```
# Input: depth, ground_mask

points_cam = unproject_to_pointcloud(depth, fov)
points = points_cam
ground_mask = ground_mask.flatten()
canonicalized = False

# Canonicalize if sufficient ground is detected
if ground_mask.mean() > canonicalize_threshold:
    canonicalized = True
    ground_pcd = subset(points_cam, ground_mask)
    # Segment ground plane
    plane, _ = ground_pcd.segment_plane(
        distance_threshold=0.05, ransac_n=3, num_iterations=1000)

    # Ensure normal points up
    if array([0, -1, 0]) @ plane[:3] < 0:
        plane = -plane

    a, b, c, d = plane
    normal = array([a, b, c])
    ez = array([0, 0, 1])

    # Calculate canonical Y-axis
    new_y = ez - normal @ ez * normal
    new_y = new_y / norm(new_y)

    # Construct rotation matrix
    rot = array([cross_prod(new_y, normal), new_y, normal])

    # Apply additional rotation
    rot = array([[0, -1, 0], [1, 0, 0], [0, 0, 1]]) @ rot

    # Translation
    trans = array([0, 0, d])

    # Transform points to canonical frame
    points_world = points_cam @ rot.T + trans[None]
    points = points_world

return points, canonicalized
```

Figure 11: Canonicalization Algorithm.

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